

Retail Brand Performance Analytics Dashboard using Azure SQL & Power BI

Project Overview

This project focuses on building an end-to-end Retail Brand Performance Analytics Dashboard using Azure SQL Database and Power BI. The project was designed to analyze retail brand performance, discounts, profitability, product variety, and sales trends through interactive dashboards and cloud-based analytics solutions.

The workflow includes:

- Loading retail data into Azure SQL Database
- Performing SQL-based data cleaning
- Connecting Azure SQL with Power BI
- Creating DAX calculations
- Building interactive dashboards
- Publishing reports using Power BI Service

This project demonstrates practical skills in SQL, cloud databases, Power BI, data visualization, and business analytics.

Objectives

- Analyze retail brand performance
- Identify top-performing and low-performing brands
- Measure discount and profitability trends
- Build interactive dashboards for business reporting
- Practice cloud-based analytics using Azure SQL Database
- Improve Power BI dashboarding and DAX skills

Tools & Technologies Used

- Microsoft Azure
 - Azure SQL Database
 - SQL
 - Power BI
 - DAX
 - Power BI Service
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Dataset Information

The dataset contains retail product information related to men's T-shirts including:

- Brand
 - Product Name
 - Marked Price
 - Sales Price
 - Discount Information
 - Profit Metrics
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Project Workflow

1. Creating Azure SQL Database

A free Microsoft Azure account was created and an Azure SQL Database instance was configured for storing and managing retail sales data.

2. Loading Data into Azure SQL Database

Retail datasets were uploaded into Azure SQL Database for cloud-based analytics and reporting.

SQL Data Cleaning

Data cleaning and preprocessing were performed directly in Azure SQL Database using SQL queries before connecting the dataset to Power BI.

The cleaning process focused on:

- Removing unwanted symbols from pricing columns
- Standardizing data formats
- Improving data consistency
- Preparing clean data for dashboard reporting and calculations

SQL Queries Used

Cleaning Original Price Column

```
SELECT TOP (1000) *  
FROM [dbo].[Men+Tshirt (1)]  
  
UPDATE [dbo].[Men+Tshirt (1)]  
SET original_price =  
TRIM(REPLACE(CAST(original_price AS VARCHAR(MAX)), '?', ''))  
WHERE original_price LIKE '%?%'
```

Cleaning Sales Price Column

```
UPDATE [dbo].[Men+Tshirt (1)]
SET sale_price =
TRIM(REPLACE(CAST(sale_price AS VARCHAR(MAX)), '?', ''))
WHERE sale_price LIKE '%?%'
```

Connecting Power BI with Azure SQL Database

Power BI was connected with Azure SQL Database using:

- Database authentication
- Microsoft account authentication

This enabled cloud-based reporting and dashboard analysis.

Data Transformation in Power BI

Additional data cleaning and formatting were performed in Power BI Power Query to improve dashboard usability and reporting quality.

DAX Calculations

Several DAX measures and calculated columns were created to analyze profitability and discount trends.

Discount Percentage

Discount % =
DIVIDE(
 'Men+Tshirt (1)'[Marked price] -
 'Men+Tshirt (1)'[Sales_Price],
 'Men+Tshirt (1)'[Marked price]
) * 100

Purpose:

- Calculates product discount percentage
 - Helps analyze heavily discounted brands
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Cost Price

Cost price =
DIVIDE(
 100 * 'Men+Tshirt (1)'[Sales_Price],
 100 + 'Men+Tshirt (1)'[Profit %]
)

Purpose:

- Estimates product cost price
 - Supports profitability analysis
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Profit Percentage

Profit % = RANDBETWEEN(2,17)

Purpose:

- Generates profit percentage values for analysis and visualization
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Dashboard Development

Brands Overview Dashboard

An interactive dashboard was created in Power BI to analyze brand performance using multiple visualizations.

Dashboard Visuals

- Bar Chart – Top 5 Brands by Highest Average Discount %
 - Donut Chart – Top 5 Brands by Variety Count
 - Ribbon Chart – Top 5 Brands by Average Sales Price
 - Area Chart – Top 5 Brands by Highest Average Profit %
 - Circle Graph – Bottom 5 Brands by Profit %
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Key Insights Generated

- Identified brands offering the highest discounts
 - Analyzed profitable and low-performing brands
 - Compared average sales prices across brands
 - Evaluated product variety distribution
 - Generated business insights using interactive visualizations
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Business Impact

The dashboard helps businesses:

- Monitor brand performance
- Improve pricing strategies

- Identify profit optimization opportunities
 - Support data-driven business decisions
 - Track low-performing products and brands
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Power BI Service Publishing

The completed dashboard was published to Power BI Service for cloud accessibility, dashboard sharing, and report management.

Skills Demonstrated

- Azure SQL Database
 - SQL Query Writing
 - Data Cleaning
 - Power BI Dashboard Development
 - DAX Calculations
 - Cloud Data Integration
 - Data Visualization
 - KPI Reporting
 - Business Analytics
 - Power BI Service Publishing
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Conclusion

This project demonstrates an end-to-end cloud-based analytics workflow using Azure SQL Database and Power BI. The dashboard provides actionable insights into retail brand performance, discounts, pricing, and profitability through interactive business intelligence reporting and visualization techniques.